C4.7 Make transactions

This section defines the state changes associated with the Make transaction group that are issued on the AR channel. The Make transactions are:

- *MakeUnique*
- *MakeInvalid* on page C4-201.

C4.7.1 MakeUnique

A MakeUnique transaction is used in a region of memory that is shareable with other masters. The MakeUnique transaction ensures that:

- The cache line can be held in a Unique state. This permits the master to carry out a store operation to the cache line, but the transaction does not obtain a copy of the data for the master.
- All other copies of the cache line are removed.

Note

A MakeUnique transaction must only be used by an initiating master that is carrying out a full cache line store operation.

The transaction response requirements are:

- the IsShared response must be deasserted indicating that the cache line is unique
- the PassDirty response must be deasserted.

The expected cache line state changes for a MakeUnique transaction are different from all other transactions because a MakeUnique transaction must be coupled to a full cache line store operation.

Table C4-17 shows the expected cache line state changes for the MakeUnique transaction with a full cache line store operation:

Table C4-17 Expected MakeUnique cache line state changes

Transaction	Start state	RRESP[3:2] IsShared/PassDirty	Expected end state	Legal end state With Snoop Filter	Legal end state No Snoop Filter
MakeUnique with full cache line store	I	00	UD	UD, SD	UD, SD
	SC	00	UD	UD, SD	UD, SD
	SD	00	UD	UD, SD	UD, SD

Table C4-18 shows the other permitted cache line state changes for the MakeUnique transaction with a full cache line store operation:

Table C4-18 Other permitted MakeUnique cache line state changes

Transaction	Start state	RRESP[3:2] IsShared/PassDirty	Expected end state	Legal end state With Snoop Filter	Legal end state No Snoop Filter
MakeUnique with full cache line store	UC	00	UD	UD, SD	UD, SD
	UD	00	UD	UD, SD	UD, SD

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C4.7.2 MakeInvalid

A MakeInvalid transaction is a broadcast cache invalidate operation. It can be used in shareable and non-shareable memory regions.

A MakeInvalid transaction is used to ensure that there are no cached copies of a main memory location.

Note

If the master carrying out the cache maintenance operation holds the cache line in a Valid state, then the master must invalidate the cache line, so that the cache line is in the Invalid state before it issues a MakeInvalid transaction.

The transaction response requirements are:

- the IsShared response must be deasserted
- the PassDirty response must be deasserted.

Table C4-19 shows the expected cache line state changes for the MakeInvalid transaction:

Table C4-19 Expected MakeInvalid cache line state changes

Transaction	Start state	RRESP[3:2]	Expected end state	Legal end state	
		IsShared/PassDirty		With Snoop Filter	No Snoop Filter
MakeInvalid	I	00	I	I	I

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